Plexa Mobility Module: Data Map for NYC DOT Traffic Volume Counts

1. Purpose

One reference for cleaning and unifying the two NYC DOT traffic volume datasets before the baseline model. All numbers come from tests run in the notebook (`hist` = historical, `a` = automated). Items marked (hypothesis) are inferences, not confirmed facts.

2. The two datasets at a glance

	Historical	Automated
NYC Open Data name	Traffic Volume Counts (Historical), static	Automated Traffic Volume Counts, current feed
Shape	Wide: one row per segment, direction, date; 24 hourly columns	Long: one row per segment, direction, timestamp
Rows x columns	42,756 x 31	1,875,154 x 14
Date range	2012-01-08 to 2021-05-09	2000-01-01 to 2026-02-03 (bulk 2009 to 2025)
Resolution	Hourly (clock hour)	Mostly 15-minute; some hourly; one odd-minute study in 2022
Segments	1,956 SegmentIDs	3,391 SegmentIDs
Geometry and borough	None	WKT POINT (EPSG:2263) and Boro
Typical coverage	Median 9 dates per segment; 52% of segments have one year only; 73% of campaigns last 9 days	Median 4 dates per segment; 932 segments have a single day; 89% have one year only
Months covered	Sep to Nov 84%, Jan to May 16%, Dec 17 rows, none in Jun to Aug	All 12 months

	Historical	Automated
Days of week	Weekend-heavy (about 8.8k rows each for Sat and Sun vs about 5k per weekday)	Balanced

3. Source columns and what they mean

Historical

Column	Meaning
ID	Count ID (text). Not a row key or segment key: 377 IDs map to more than one SegmentID
SegmentID	LION Segment ID. NYC documentation says it was called GIS ID in the 2011-12 dataset and Segment ID in 2012-13
Roadway Name, From, To	Free text with mixed case and abbreviations. Dead End appears as a placeholder in From and To
Direction	14 values: NB, SB, EB, WB plus -MAIN, -SER, -SR, -MR and spacing variants (212 rows carry a suffix)
Date	Text, MM/DD/YYYY
24 hour columns	Count for the clock hour. Map by position to h00 to h23. The label 11:00-12:00PM is hour 11 and 11:00-12:00AM is hour 23. Stored as text with commas (97,343 cells)

Automated

Column	Meaning
RequestID	Study or request ID. Not unique per segment: 585 RequestIDs span more than one SegmentID
Boro	Borough name (5 boroughs)

Column	Meaning
Yr, M, D, HH, MM	Timestamp of the interval START. Full days run from 00:00 to 23:45
Vol	Count per interval. Interval length varies by study (15-minute, hourly, odd-minute). Stored as text with commas (15,257 cells)
SegmentID	LION Segment ID
WktGeom	WKT POINT in EPSG:2263 (NY State Plane, feet). All 3,391 segment points fall inside NYC after conversion
street, fromSt, toSt	Free text. Older rows embed direction and intersection (for example S/B LUTEN AVE @ EYLANDT ST). toSt has 1,246 nulls
Direction	NB, SB, EB, WB plus EW (3,564 rows) and NS (2,964 rows)

The column descriptions for the automated dataset on NYC Open Data have not been read yet.

4. Target schema (canonical hourly table)

One row per segment_id, direction, hour_start.

Column	Meaning
segment_id	LION Segment ID as given in the source
direction	NB, SB, EB or WB
road_type	main, service or null (from historical suffixes -MAIN, -SER, -SR, -MR)
hour_start	Timezone-aware timestamp (America/New_York) at the start of the clock hour
volume	Vehicles in that hour; missing when not reliable
source	historical or automated

Column	Meaning
orig_interval_min	60 or 15
flags	partial_day, zero_to_missing, name_mismatch, key_conflict, covid, sparse_year
lion_key_used	SegmentID, LegacyID or none

A separate segment table holds segment_id, the LION attributes listed in section 8, and the geometry. Raw source tables stay untouched.

5. Settled facts

- Automated timestamps mark the start of the interval, so summing the four 15-minute rows of an hour gives the same clock hour as a historical column.
- Where the datasets overlap they agree: on 22,275 matched segment-direction-hours the ratio of automated (summed to hourly) to historical has median 1.0, correlation 0.9988, and 98.9% are within 10%.
- The overlap is small: 346 shared SegmentIDs and 788 shared segment-days, so the datasets mostly cover different locations.
- Coordinates are EPSG:2263. Converted to lat/lon, every point falls inside NYC and the borough medians are sensible.
- SegmentID is the LION Segment ID. The street centerline file (CSCL) uses a different ID (PhysicalID) and does not match.

6. Cleaning rules

1. Types. Strip commas, then convert hour columns and Vol to numbers. Parse Date as MM/DD/YYYY. The 8 non-integer historical cells can be rounded or kept as decimals.
2. Historical hours. Rename by position (h00 to h23), then melt to long.
3. Timestamps. Build one timezone-aware timestamp (America/New_York). Flag segment-days that look like daylight saving transitions (not yet tested).
4. Resolution. For each study-day (RequestID, SegmentID, Direction, date) infer the interval from the set of minute values. 15-minute: sum the four intervals of an hour only if all four are present, otherwise leave the hour missing. Hourly studies (minute set is only 0): keep as is. Odd-minute rows (10, 20, 40, 50): quarantine.

5. Direction. Keep NB, SB, EB, WB as direction. Split historical suffixes into road_type (the meaning of SR and MR is a guess; check against names). Send EW and NS to quarantine.
6. Zeros, historical. Set to missing: trailing zeros when the run is 3 hours or longer. Trailing runs of 1 to 2 hours and isolated interior zeros are set to missing only when the segment-hour median is above 100 (34 isolated interior cases).
7. Zeros, automated. Set to missing: runs of zeros lasting 2 hours or more (72.5% of zero cells sit in runs of 8 or more intervals). Keep shorter runs. Set the single -1 value to missing.
8. Partial days. Keep valid hours and flag the day. Use daily totals only from complete days.
9. Duplicates. Quarantine duplicate keys in both datasets. Never sum or average them.
10. Overlap between datasets. Where the same segment, direction, date and hour exists in both, keep the automated value and drop the historical copy.
11. Names. Build name_norm for validation only, never as a key. Expand abbreviations (PKWY, FT, MT, HWY, TER), strip ordinals (3RD to 3), W/E/N/S prefixes, punctuation and spacing, strip embedded direction and AT/@ text in automated names, and use an alias list (6 AVENUE and AVENUE OF THE AMERICAS, FDR DRIVE and FRANKLIN D ROOSEVELT DRIVE).
12. Flags. partial_day, zero_to_missing, name_mismatch, key_conflict, covid (from 2020-03; end date decided at modeling time), sparse_year.

7. Quarantine tables

Table	Rows	Reason
hist_duplicates	350 (175 pairs)	Same SegmentID, Direction and Date. 113 pairs differ in raw names and volume ratios reach 90x, so many are different locations sharing an ID. Concentrated in 2012, 2013 and 2015
auto_duplicates	10,292 (5,146 keys)	All in 2023 and 2024, from 3 RequestIDs and 6 segments. The two copies are different series (correlation 0.85)

Table	Rows	Reason
auto_twoway	6,528 (EW 3,564; NS 2,964)	Meaning unknown. On the one segment that could be compared, NS was about 5% of NB plus SB, so they are not simple two-way totals
auto_oddminute	2,871	One RequestID, 26 segments, 4 readings per hour at minutes 10, 20, 40, 50 (not true 10-minute counts). No overlapping study to check against
no_lion_match	495 historical rows (36 segments); 74 automated segments	No LION record found by SegmentID or LegacyID

8. Joining to LION (geometry and segment attributes)

LION (NYC Department of City Planning) loaded from the file geodatabase: layer lion, 243,237 rows x 129 columns, EPSG:2263.

Match order: 1) LION SegmentID; 2) LegacyID if the SegmentID is not found; 3) otherwise no match. Store lion_key_used.

Result	Historical	Automated
Matched by SegmentID	1,716 of 1,956 (87.7%)	2,893 of 3,391 (85.3%)
Matched by LegacyID	1,724 (88.1%)	3,167 (93.4%)
Matched by either	1,920 (98.2%)	3,317 (97.8%)
Names agree, via SegmentID	1,612 of 1,716 (93.9%)	2,441 of 2,893 (84.4%)
Names agree, via either key	1,800 of 1,920	2,751 of 3,317
No match	36 segments	74 segments

- Spatial check (automated, SegmentID matches, 2,893 segments): the distance from the count point to the LION line has median 0.007 ft, 95th percentile 2.6 ft, 99th percentile 11.9 ft and maximum 127 ft. Almost all are within 100 ft, so the automated name disagreements are mostly formatting, not wrong IDs.

- Historical has no geometry, so only the name check applies. Of 15 mismatches shown, most were formatting or aliases; a few look genuine (SegmentIDs 202, 1884, 1886, and possibly 23942). Flag with name_mismatch and review; do not drop.
- Name agreement does not vary much by the segment's first year (2012: 95.4%; other years 90.5% to 100%), so the idea that 2012 IDs are less reliable is not supported.
- 2,000 historical rows have SegmentID above 300,000 (for example 9001519 and 8103575). They are ordinary roads (parkways, Fordham Road, White Plains Road), and about 89% of these IDs exist in LION.
- LION attributes to attach: POSTED_SPEED, Number_Travel_Lanes, Number_Park_Lanes, Number_Total_Lanes, StreetWidth_Min, StreetWidth_Max, StreetWidth_Irr, FCC, TrafDir, TRUCK_ROUTE_TYPE, Snow_Priority, BikeLane, LBoro, RBoro, SHAPE_Length, geometry.

9. Cautions for modeling

- Seasonality: historical has no June to August data and 84% of it is September to November, so a baseline from historical alone describes autumn. Automated covers all months and should provide seasonal factors.
- Thin coverage: median 9 dates (historical) and 4 dates (automated) per segment, and most segments are seen in a single year, so per-segment seasonality cannot be learned for most of them.
- Evaluation: a random split leaks across a campaign. Hold out days within a campaign and, separately, whole segments. Use a time split only for multi-year segments.
- Weekend overrepresentation in historical: stratify by day of week.
- Scale: hourly volumes are heavy-tailed (historical median 248, maximum 10,532). Use log or normalized metrics.
- Regime and sparse years: 2020 shows a COVID dip (automated 40,224 rows; historical 5,643). Sparse years: automated 2000, 2006 to 2008 and 2026; historical 2013, 2018 and 2021.
- The automated zero share is higher in 2008 to 2010 (9% to 12%) than from 2020 onward (1% to 3%).

10. Open items

1. Check whether segments matched by both SegmentID and LegacyID point to different LION rows (key_conflict).

2. The meaning of EW and NS, and of the odd-minute study, is still unexplained.
3. The automated column descriptions on NYC Open Data have not been read.
4. Daylight saving handling is untested.
5. Whether low SegmentIDs (202, 1884, 1886) are less reliable is untested.
6. The alias list for street names is not built yet.
7. The unit of Vol in hourly studies is assumed to be vehicles per hour.

11. Reconciliation checklist

At every step, rows in must equal rows out plus quarantined rows. These counts should reproduce:

- Historical: 42,756 rows; 1,023,064 non-null hourly cells before zero rules; 257 partial-day rows.
- Automated: 1,875,154 rows; 5,146 duplicate keys; 96,278 zero cells; one negative value.
- Overlap: 788 shared segment-days; 346 shared SegmentIDs.
- LION: 243,237 rows loaded; 1,920 historical and 3,317 automated segments matched.